KiPI Coupled Noise report: +3.3V

Audio_Visualiser_DSP.kicad_pcb · 2026-10-19 01:01 · KiPI 0.9.0

Overall: **PASS** (10 of 10 victim(s) within their noise margin).

Victims

Victim	Net	Driver → receiver	Peak (mV)	Margin (mV)	Of margin	By mechanism (mV)	Dominant	Result
Net-(U402-PG) (scan)	Net-(U402-PG)	U401[6] → U402[4]	11.7	400	2.92 %	capacitive 1.26, inductive 12.8, common 0.192	inductive	**PASS**
I2C_SDA (scan)	I2C_SDA	U1002[1] → U302[1]	0.0003171	400	7.93e-05 %	capacitive 0, inductive 0.000317, common 0	inductive	**PASS**
I2C_SCL (scan)	I2C_SCL	U1001[5] → U302[2]	0.0001093	400	2.73e-05 %	capacitive 0, inductive 0.000109, common 0	inductive	**PASS**
POS3P3_RUN (scan)	POS3P3_RUN	U402[13] → Q401[1]	0.3076	400	0.0769 %	capacitive 0.00755, inductive 0.313, common 0	inductive	**PASS**
Net-(U402-SS{slash}TR) (scan)	Net-(U402-SS{slash}TR)	U402[9] → C404[1]	0.7272	400	0.182 %	capacitive 0.000291, inductive 0.606, common 0.213	inductive	**PASS**
Net-(U402-FB) (scan)	Net-(U402-FB)	U402[5] → R405[1]	1.725	400	0.431 %	capacitive 0.0381, inductive 1.75, common 0.192	inductive	**PASS**
POS12_SNS- (scan)	POS12_SNS-	U301[D2] → C301[2]	2.336	400	0.584 %	capacitive 0, inductive 2.34	inductive	**PASS**
POS12_PGOOD (scan)	POS12_PGOOD	U701[26] → R208[2]	0.0006084	400	0.000152 %	capacitive 0, inductive 0.000608	inductive	**PASS**
Net-(D401-K) (scan)	Net-(D401-K)	D401[1] → C408[1]	0.1118	400	0.028 %	capacitive 0, inductive 0.112	inductive	**PASS**
Net-(Q401-D) (scan)	Net-(Q401-D)	Q402[1] → Q401[3]	0.05354	400	0.0134 %	capacitive 0, inductive 0.00493, common 0.0511	common	**PASS**

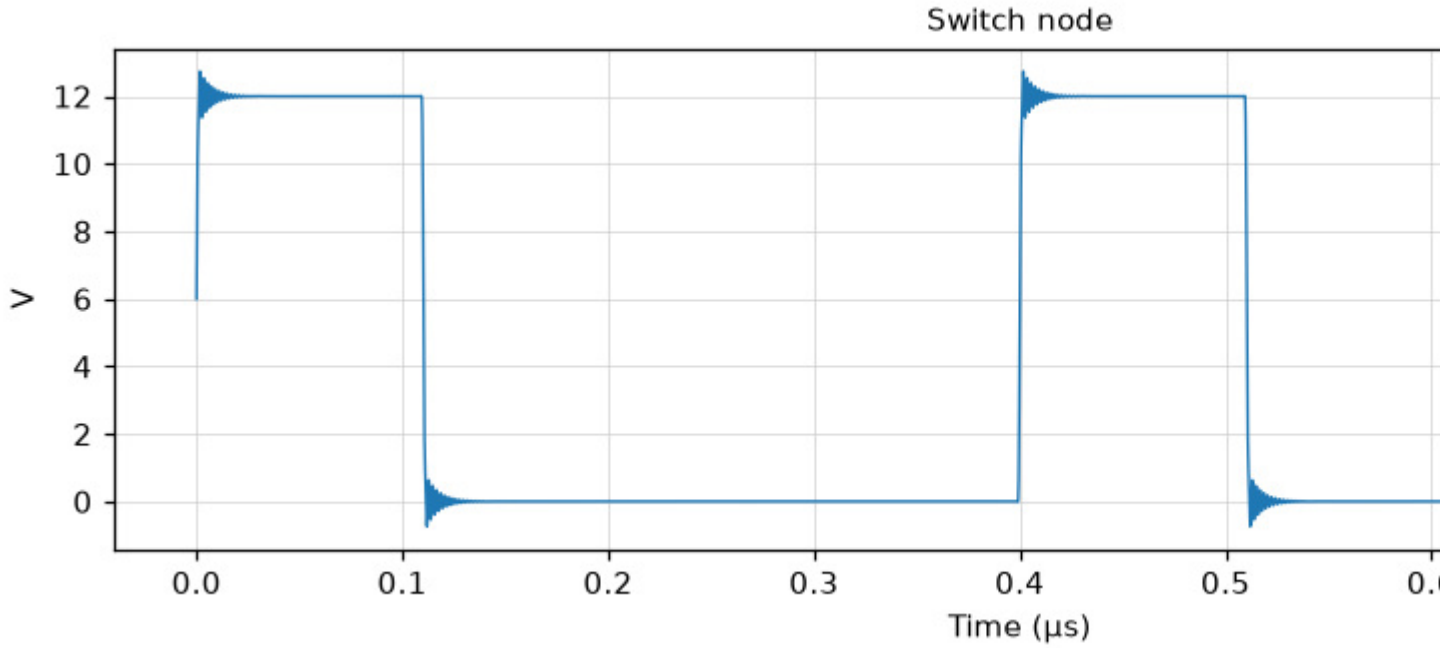
Peak: the largest noise at the receiver over a switching period, all mechanisms added as phasors. Scanned nets use the scan's default driver and logic level.

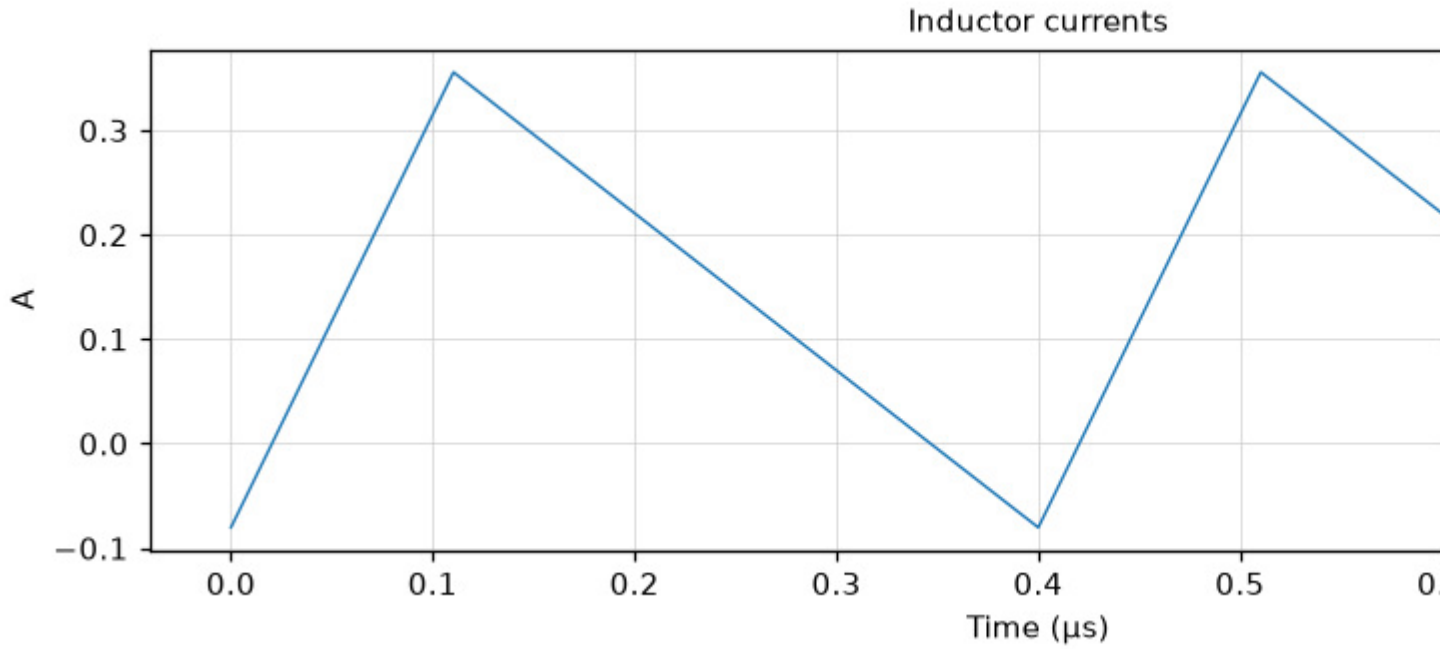
Aggressor

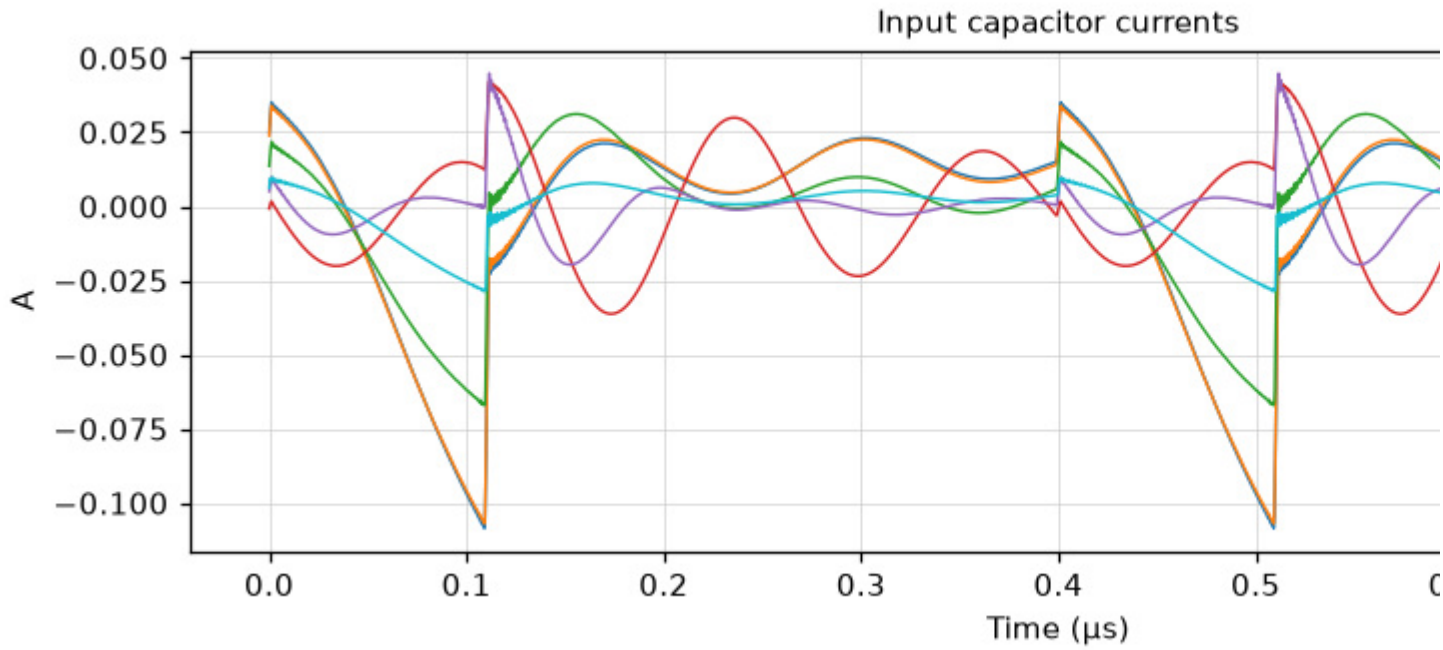
- Topology: buck (from the voltages)
- Vin 12 V (middle of +3.3V's input range), Vout 3.3 V (+3.3V)
- Iout 0.1377 A (DC setup's loads on Net-(C406-Pad1))
- Duty 0.2756
- Inductor ripple 0.436 A p-p; harmonics to 1608 MHz (643)
- Phase 1: hot loop 0.436 nH (package 0 nH), switch node 202.5 pF to ground (Coss 200 pF assumed + copper 2.5 pF), R 96.1 mΩ; rings at 535.7 MHz, ζ = 0.05 (least damping; the loop alone gives 0.0328)
- Board model: 80 capacitor(s) on the rails, 2284644 unknowns, 30 field solves, plane capacitance 4.86 nF
- Scan: 29 net(s) near the aggressor estimated, the worst 10 solved in full

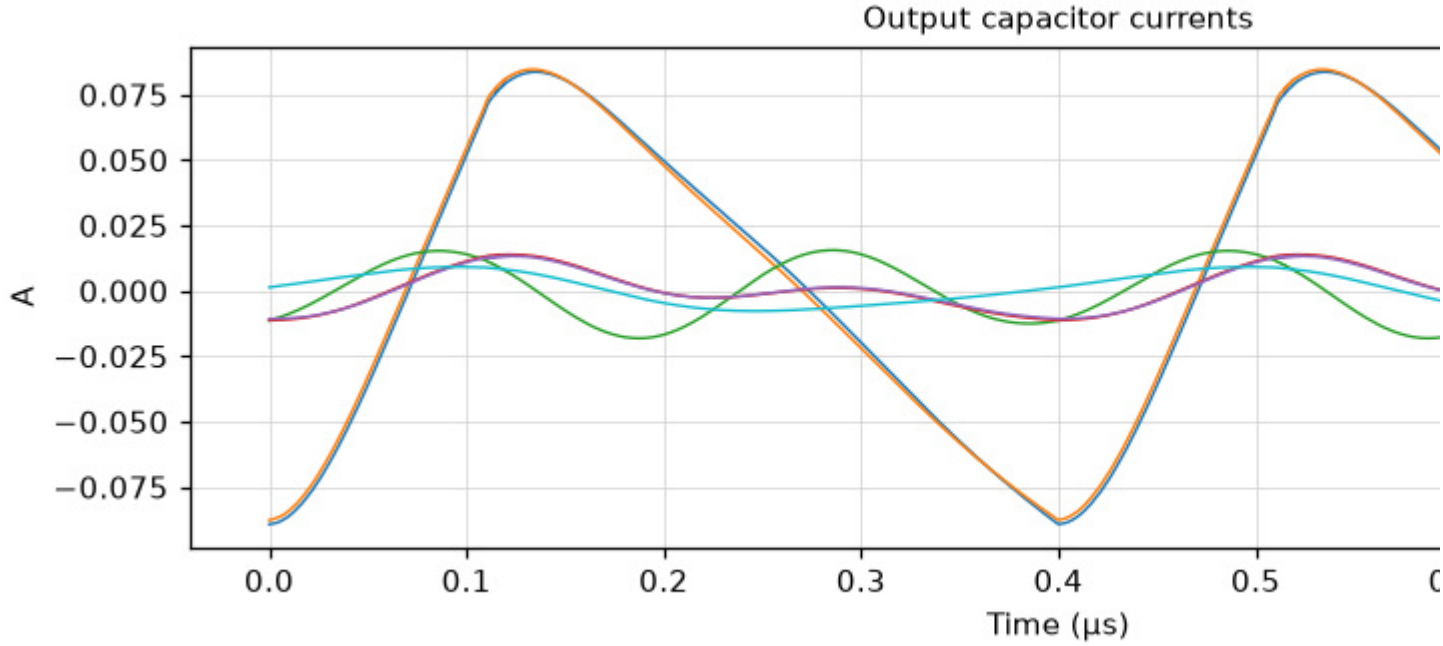
Phase	Hot loop (nH)	Switch node C (pF)	Coss (pF)	Copper C (pF)	Loop R (mΩ)	Ringing (MHz)	ζ
1	0.436	202.5	200 (assumed)	2.5	96.1	535.7	0.05

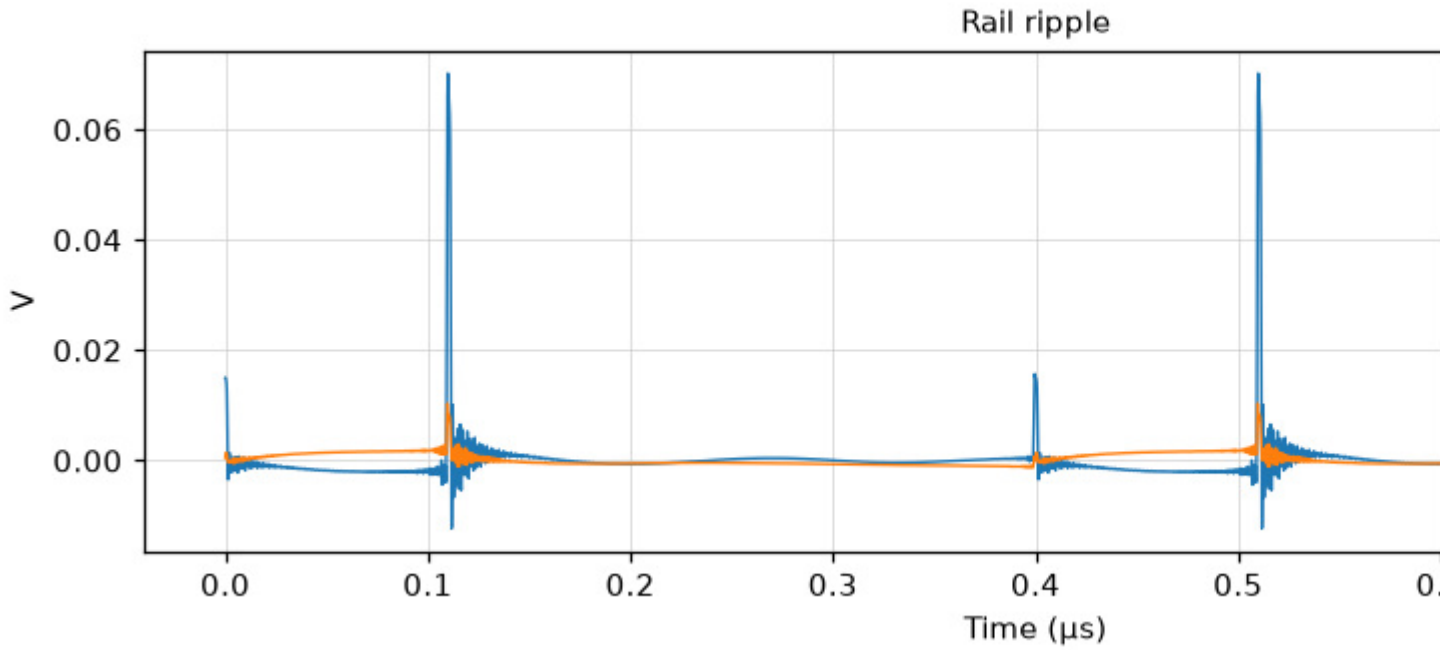
Waveforms


Switch node
12
10
8
6
4
2
0
V
0.0
0.1
0.2
0.3
0.4
0.5
Time (µs)


Inductor currents
0.3
0.2
0.1
0.0
−0.1
A
0.0
0.1
0.2
0.3
0.4
0.5
Time (µs)

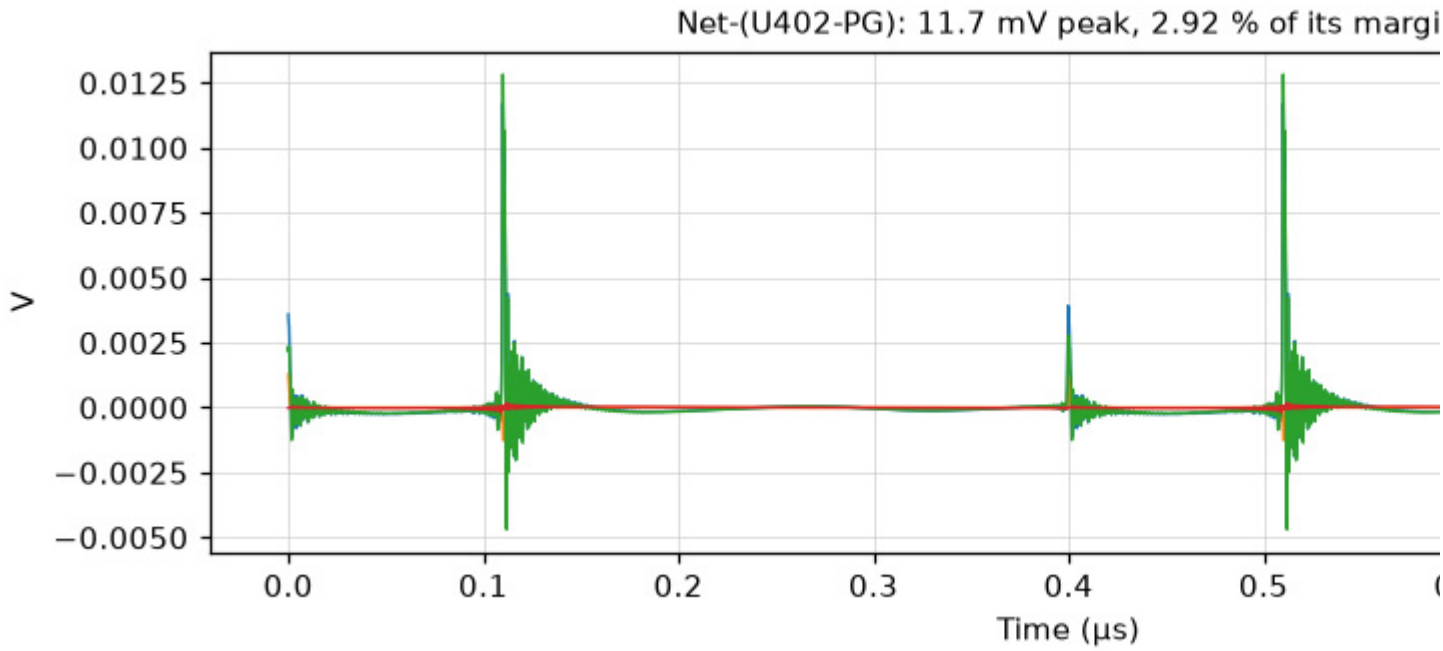

Input capacitor currents
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
A
0.0
0.1
0.2
0.3
0.4
0.5
Time (µs)


Output capacitor currents
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
A
0.0
0.1
0.2
0.3
0.4
0.5
Time (µs)

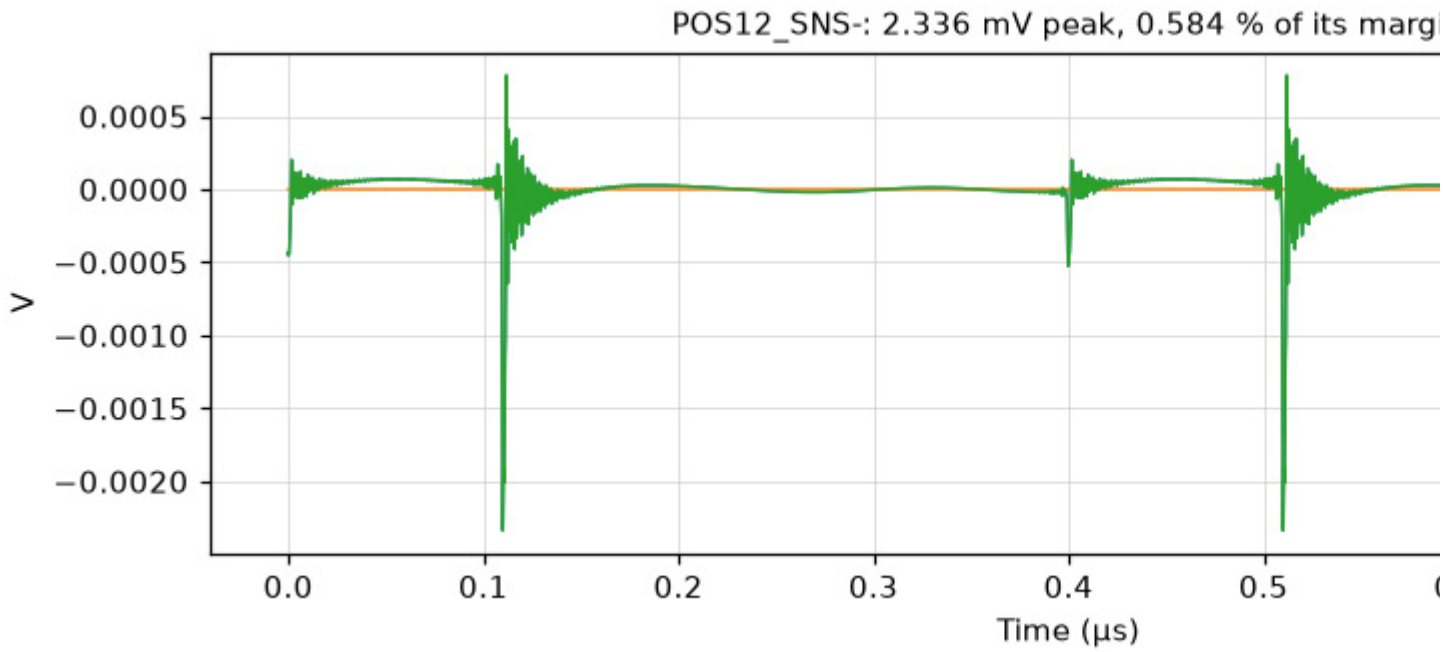


Victim noise

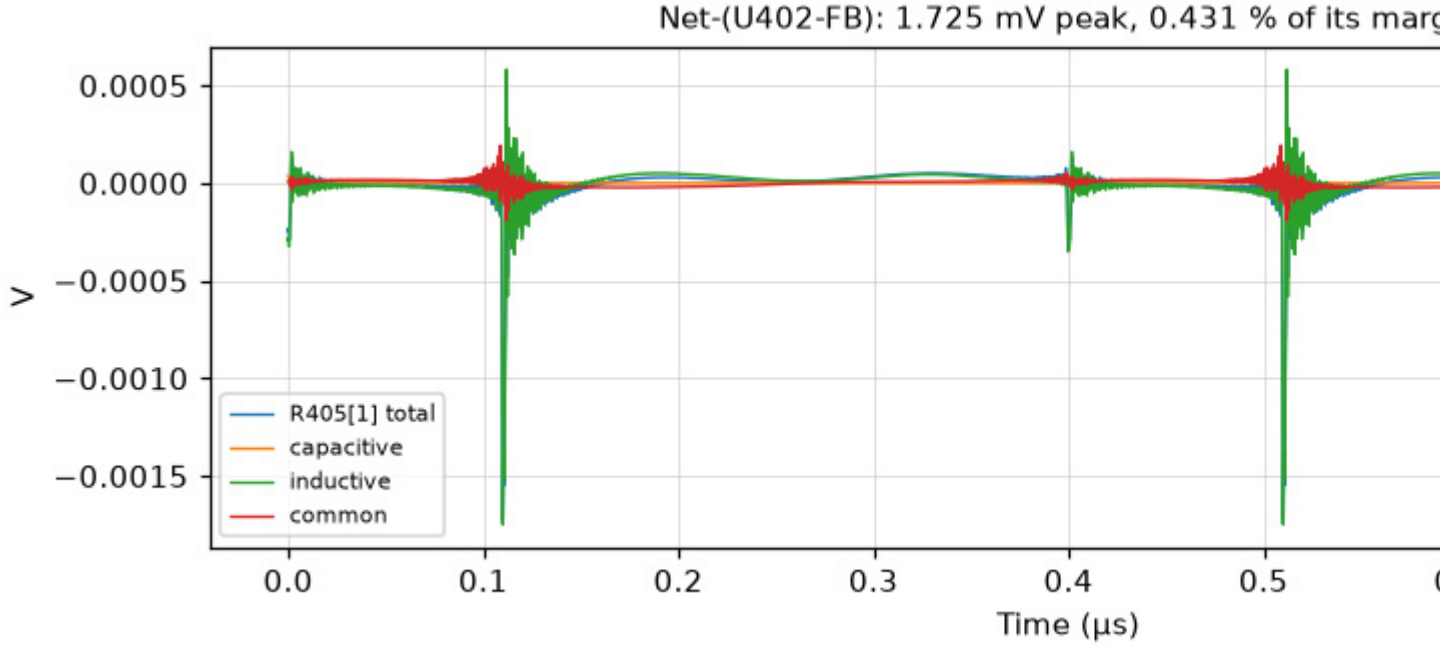
Net-(U402-PG)



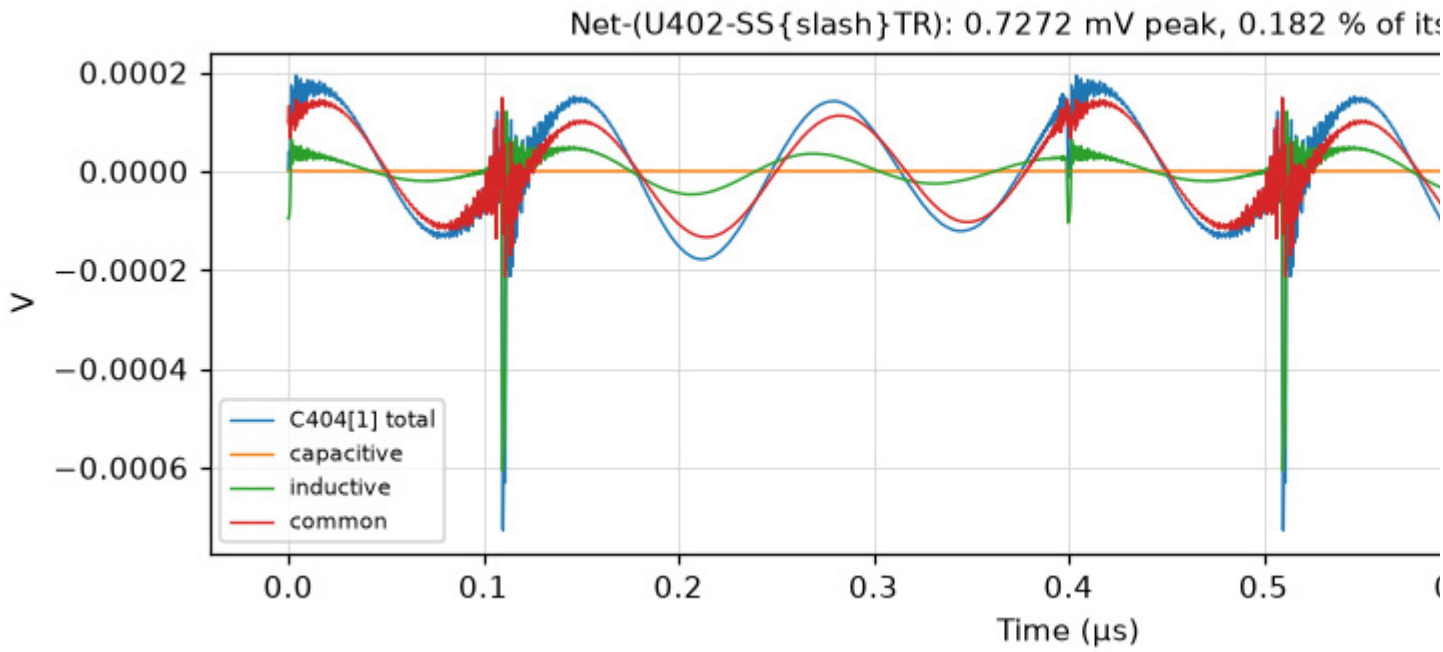
POS12_SNS-



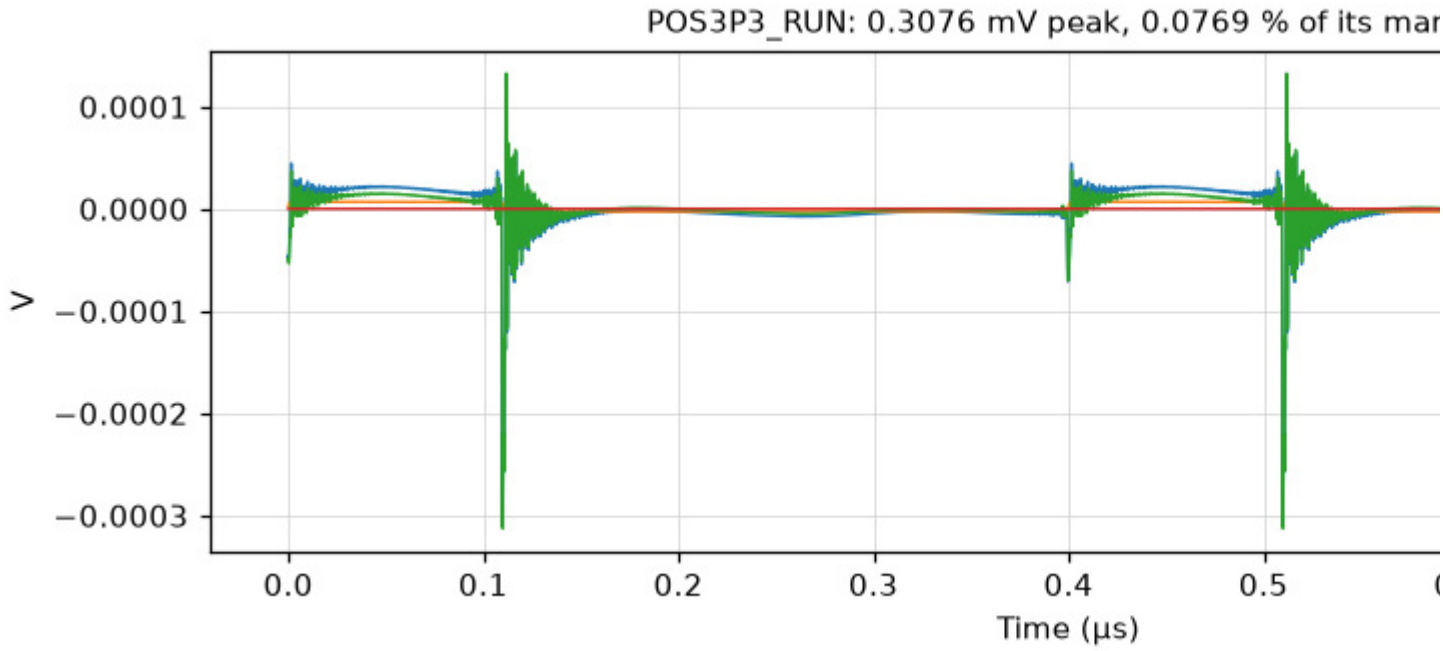
Net-(U402-FB)



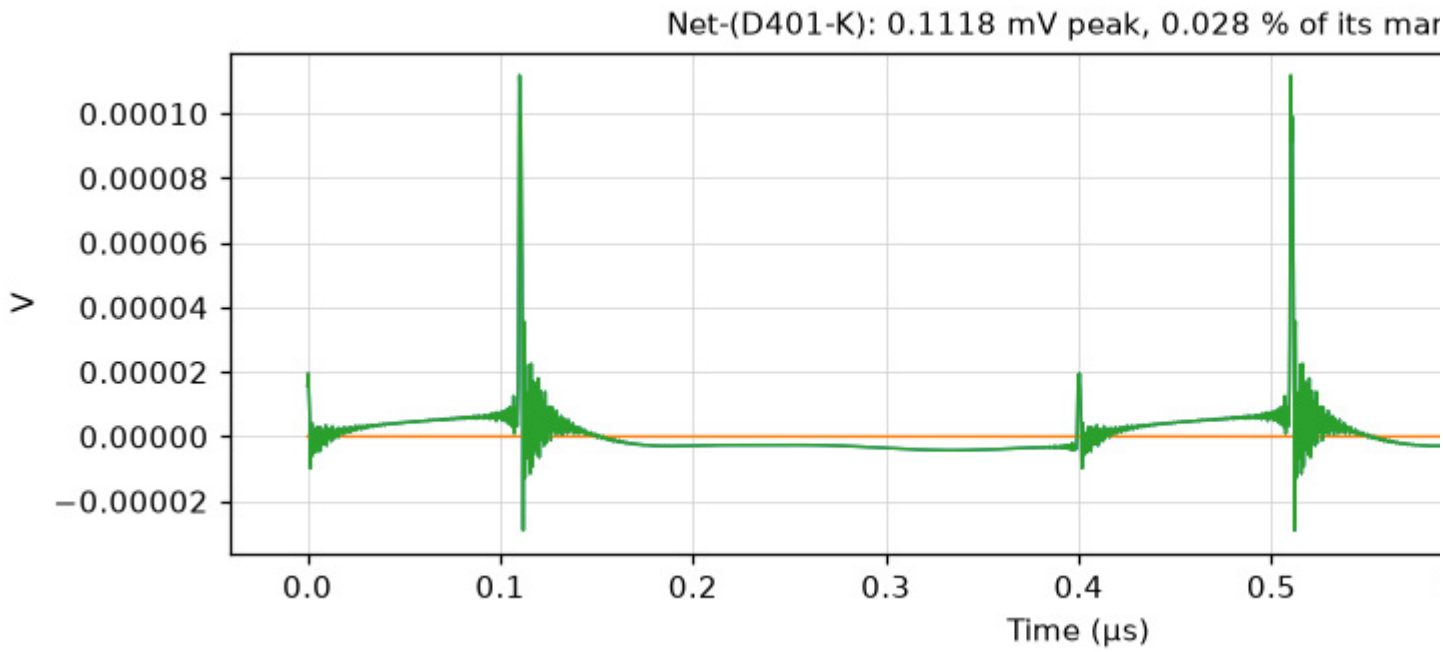
Net-(U402-SS{slash}TR)



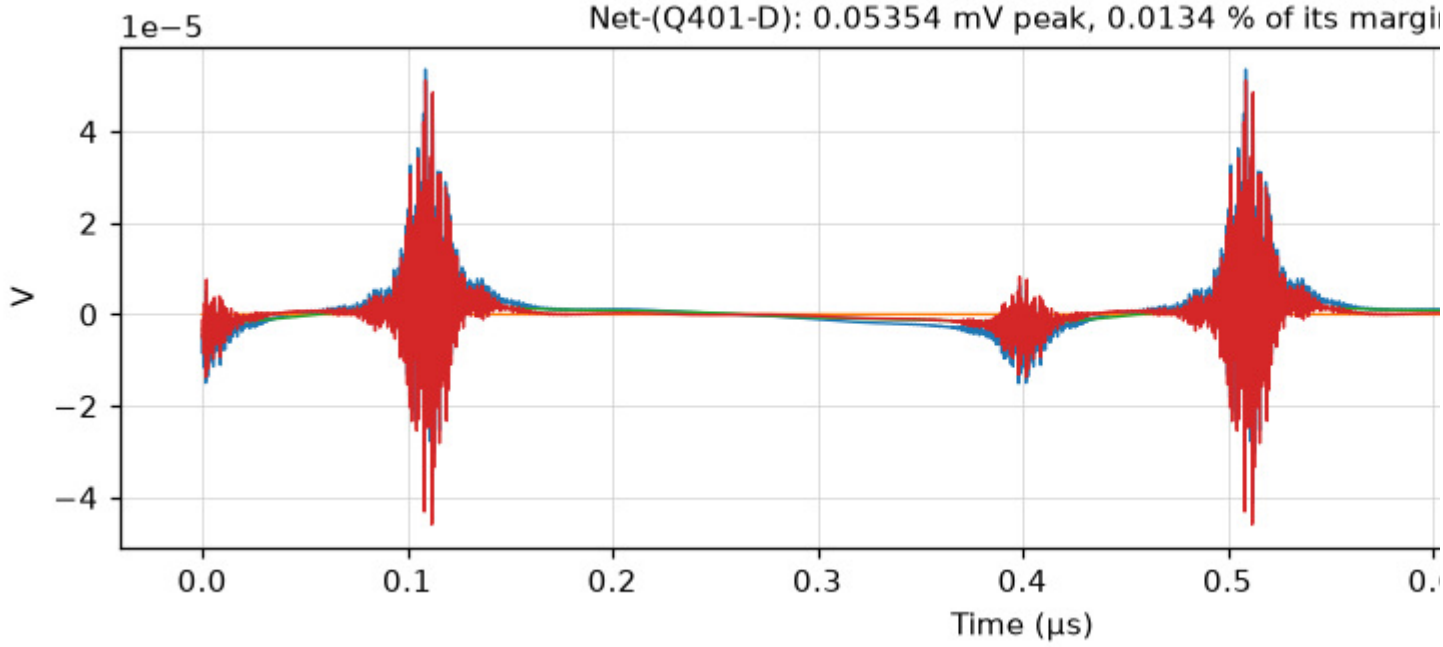
POS3P3_RUN



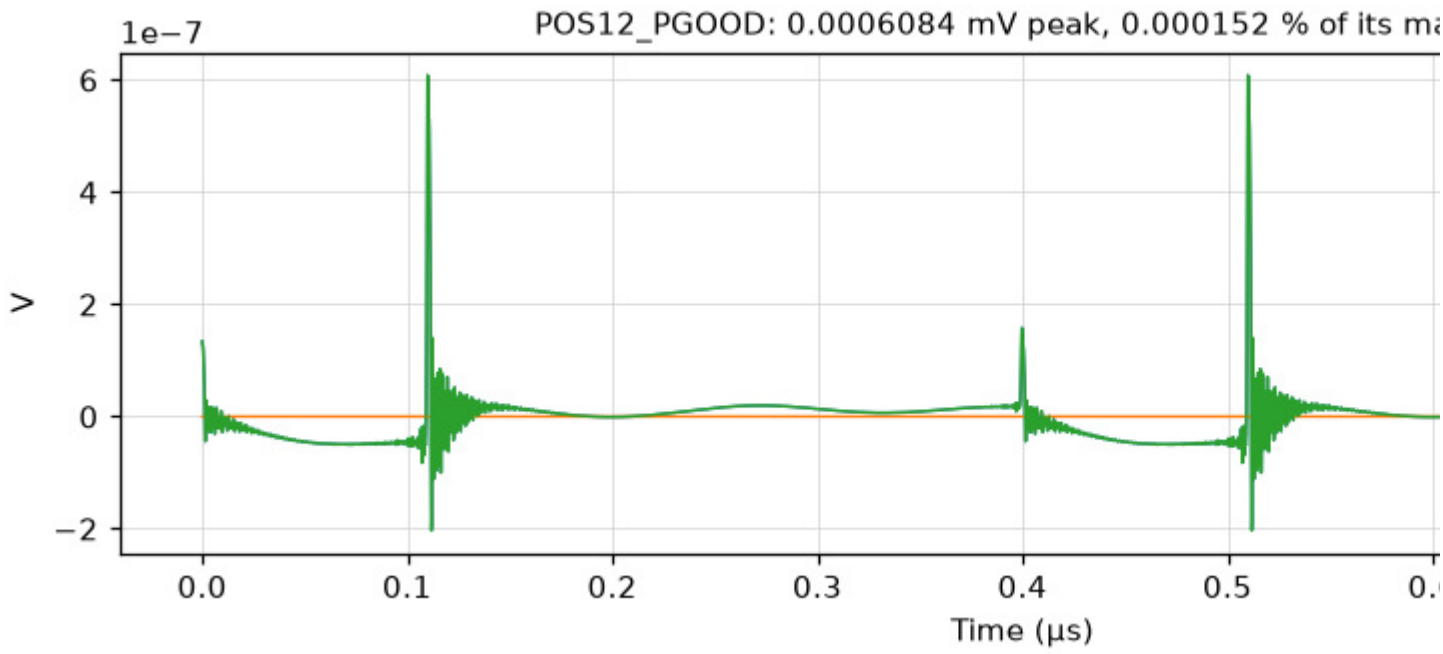
Net-(D401-K)



Net-(Q401-D)



POS12_PGOOD



The 8 victims nearest their margin are plotted.

Scan

Net	Distance (mm)	Estimate (mV)	Capacitive (mV)	Inductive (mV)	Solved in full
Net-(U402-PG)	0	7.696	5.41	2.286	yes
I2C_SDA	0	5.485	4.015	1.47	yes
I2C_SCL	0	4.271	2.879	1.392	yes
POS3P3_RUN	0	3.292	2.758	0.5344	yes
Net-(U402-SS{slash}TR)	0	2.929	0.1406	2.789	yes
Net-(U402-FB)	0	0.9667	0.7964	0.1702	yes
POS12_SNS-	0.809	0.7171	0	0.7171	yes
POS12_PGOOD	0	0.6321	0.241	0.3911	yes
Net-(D401-K)	2.05	0.3616	0	0.3616	yes
Net-(Q401-D)	0.997	0.3609	0	0.3609	yes
POS3P3_PGOOD	0	0.2069	0.1107	0.09619	
~{USB_BRIDGE_RESET}	9.19	0.07837	0	0.07837	
USB_UART_RX	9.38	0.0436	0	0.0436	
POS12_SNS+	1.12	0.04305	0	0.04305	
Net-(Q402-D)	2.19	0.03578	0	0.03578	
/USB UART Bridge/USB_BRIDGE_D-	9.15	0.02235	0	0.02235	
Net-(D901-A)	9.65	0.01196	0	0.01196	
POS3P3_SNS+	2	0.005977	0	0.005977	
Net-(U201-PGTH)	6.55	0.004866	0	0.004866	
/USB UART Bridge/USB_BRIDGE_D+	9.49	0.002518	0	0.002518	
POS3P3_SNS-	2.76	0.002159	0	0.002159	
Net-(U201-ITIMER)	5.81	0.001432	0	0.001432	
Net-(U201-dVdT)	4.39	0.001384	0	0.001384	
Net-(U201-OVLO)	7.32	0.00132	0	0.00132	
Net-(FB901-Pad2)	9.47	0.001179	0	0.001179	
Net-(U201-ILM)	5.03	0.001168	0	0.001168	

Net	Distance (mm)	Estimate (mV)	Capacitive (mV)	Inductive (mV)	Solved in full
Net-(U201-EN{slash}UVLO)	7.53	0.001147	0	0.001147	
~{USB_BRIDGE_RX}	9.65	0.0009077	0	0.0009077	
Net-(U904-EN)	6.22	0.0005958	0	0.0005958	

The estimate ranks the nets near the aggressor quickly; the worst are solved in full (above, marked scan).

Notes

- Phase 1: No Coss entered: 200 pF assumed; the ringing is approximate
- Victim POS12_SNS-: its driver or receiver has no pad on GND: no shared-ground coupling
- Victim POS12_PGOOD: its driver or receiver has no pad on GND: no shared-ground coupling
- Victim Net-(D401-K): its driver or receiver has no pad on GND: no shared-ground coupling
- The board model is quasi-static: above about 1 GHz (here to 1.61 GHz) it is approximate

User notes